

Itamar Avitan

PhD candidate · NeuroAI and computational cognitive neuroscience

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github.com/Itamar-Avitan | linkedin.com/in/itamar-avitan | scholar.google.com/citations?user=3fynQuAAAAAJ

RESEARCH INTERESTS

Representational alignment between deep neural networks and human vision; model recovery and identifiability in model comparison; evaluation metrics for model–brain and model–behaviour alignment.

EDUCATION

PhD candidate, direct track (integrated M.Sc.–PhD) · 2024–present

Ben-Gurion University of the Negev (BGU), Department of Industrial Engineering and Management; School of Brain Sciences and Cognition, Israel

- Advisor: Dr. Tal Golan, Brains and Machines Lab. Expected completion: 2029.
- Admitted to the direct PhD track for outstanding B.Sc. graduates; PhD candidacy examination passed, 2026.

M.Sc. in Industrial Engineering and Management, completed within the direct PhD track · 2026

Ben-Gurion University of the Negev, Israel

B.Sc. in Cognitive and Brain Sciences (computational track) and B.A. in Psychology · 2020–2024

Ben-Gurion University of the Negev, Israel

PEER-REVIEWED PUBLICATION

Itamar Avitan and Tal Golan (2025). Model–Behavior Alignment under Flexible Evaluation: When the Best-Fitting Model Isn't the Right One. *Advances in Neural Information Processing Systems 38 (NeurIPS 2025)*, main conference track.

Paper: arxiv.org/abs/2510.23321 | Code: github.com/brainsandmachines/oddoneout_model_recovery

Data: doi.org/10.6084/m9.figshare.30542690.v1 | Recorded presentation: slideslive.com/39047290

TALKS AND PRESENTATIONS

Sep 2026 · **Flash talk**, Neuro-AI-Talks (NEAT) 2026, Osnabrück, Germany. “System identification perspective on representational alignment” (with Maya Zach and Tal Golan).

Dec 2025 · **Poster**, NeurIPS 2025, San Diego, USA; poster session for the paper listed above.

Aug 2025 · **Contributed talk and poster** (with Tal Golan), Cognitive Computational Neuroscience (CCN) 2025, Amsterdam. “Rethinking Representational Alignment: Linear Probing Fails to Identify the Ground-Truth Model”; preliminary version of the NeurIPS 2025 paper. Session recording: youtube.com/watch?v=vT-3kV89Rhk&t=715s

RESEARCH EXPERIENCE

PhD researcher, Brains and Machines Lab · 2024–present

Ben-Gurion University of the Negev, Israel

- Built a large-scale model-recovery test for representational-alignment metrics: linearly aligned 20 diverse vision models to 4.5 million human odd-one-out judgments (THINGS dataset) and tested whether the data-generating model is re-identified from simulated responses.
- Showed that recovery accuracy plateaus below 80% even with millions of simulated trials; regression analyses linked misidentification primarily to shifts in representational geometry induced by the linear transformation. Flexible evaluation trades identifiability for predictive accuracy.

PROJECTS

Embodied Brain Technology Practicum (joint Brown University–BGU programme) · Jul–Aug 2026

Brown University, Providence, USA

- Five-person team prototype (EarBetter): a working system that drove selective audio attenuation from physiological signals—electroencephalography (EEG) and heart-rate variability from a Muse headband, and skin conductance from a separate sensor; worked on control software and system integration.
- Code: github.com/Itamar-Avitan/earbetter-ebt-practicum

BCI4ALS: brain-computer interface (BCI) for a person with amyotrophic lateral sclerosis (ALS) · 2022–2023
BrainstormIL and IsrALS programme, Israel

- Team of five built an EEG-based P300 yes/no communication system, used live by the person it was built for: data collection, MNE-Python preprocessing, classifier search with scikit-learn, real-time prediction on OpenBCI hardware.
- Code: github.com/arbelna/BCI4ALS

FELLOWSHIPS AND AWARDS

2024–present · **Lachish Fellowship for Outstanding Doctoral Students**, Kreitman School of Advanced Graduate Studies, BGU (five-year award).

2024–present · **Research Excellence Scholarship**, School of Brain Sciences and Cognition, BGU.

2025 · **Merit-based Travel Grant**, Data Science Research Center, BGU.

2025 · **Outstanding Poster Award**, Research Day, Department of Industrial Engineering and Management, BGU.

TEACHING AND MENTORING

2023–present · **Teaching assistant**, BGU: Introduction to Cognition and Computation (2023/24, 2024/25, 2025/26); Deep Learning for Neuroscience and Cognition (spring 2026); Computational Approaches to Neuroimaging (2024).

2025–2026 · **Research mentoring**: two undergraduate research students in the lab (model–brain alignment; statistical modelling of neural spiking data).

2021–2023 · **Academic tutor**, Dean of Students Office, BGU: calculus, linear algebra and statistics.

SERVICE

2025 · Reviewer, Cognitive Computational Neuroscience (CCN) 2025, extended-abstracts track.

SKILLS AND LANGUAGES

Technical · Python, PyTorch, NumPy/SciPy/pandas, scikit-learn; Slurm cluster computing, Git, Linux.

Languages · Hebrew (native), English (fluent).

REFERENCES

Available on request.